US 20020102455A1

(19) United States

(12) Patent Application Publication (10) Pub. No.: US 2002/0102455 A1

Daroux et al. (43) Pub. Date: Aug. 1, 2002

(54) BATTERY SEPARATOR FOR LI-ION AND/OR LI-ION POLYMER BATTERY

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(21) Appl. No.: **09/730,812**

(22) Filed: **Dec. 6, 2000**

Related U.S. Application Data

(60) Provisional application No. 60/169,860, filed on Dec. 9, 1999.

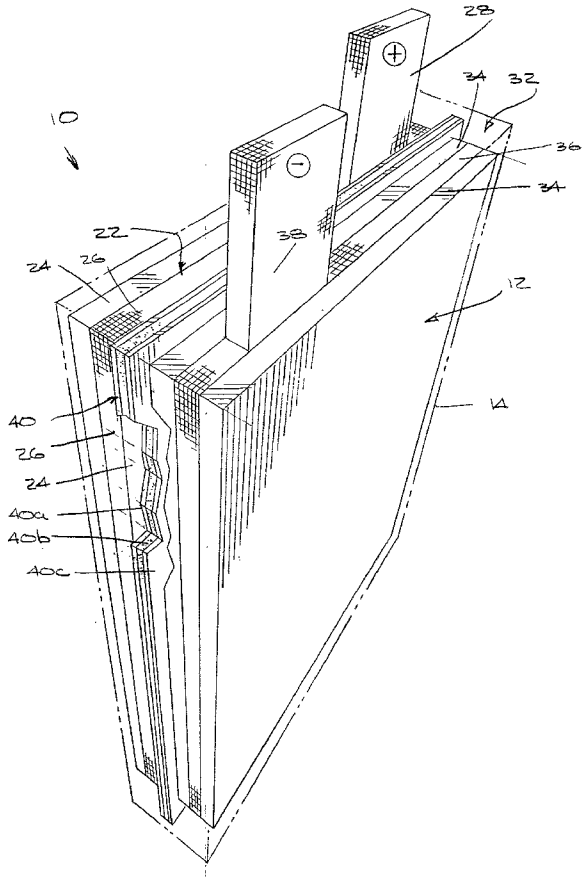
Publication Classification

(51) **Int. Cl.**[7] **H01M 2/16**; H01M 2/18

(52) **U.S. Cl.** **429/144**; 429/252; 29/623.3

(57) **ABSTRACT**

A separator for a Li-ion polymer battery comprised of a plurality of separator layers that are laminated together. The plurality of separator layers including a first layer formed of a first separator material, and a second layer formed of a second separator material, wherein the second layer is compositionally and structurally different from the first layer.



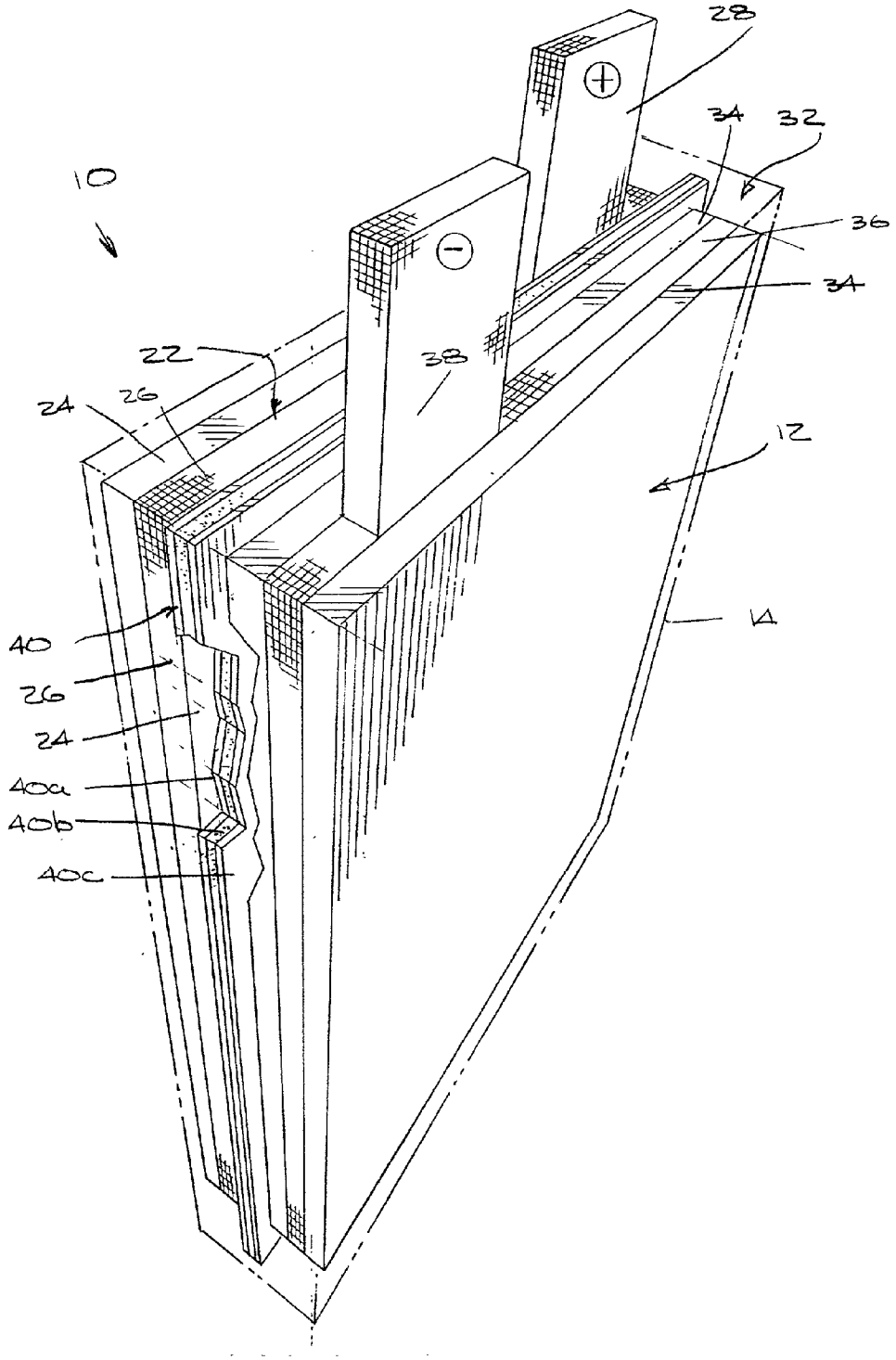


FIG. 1

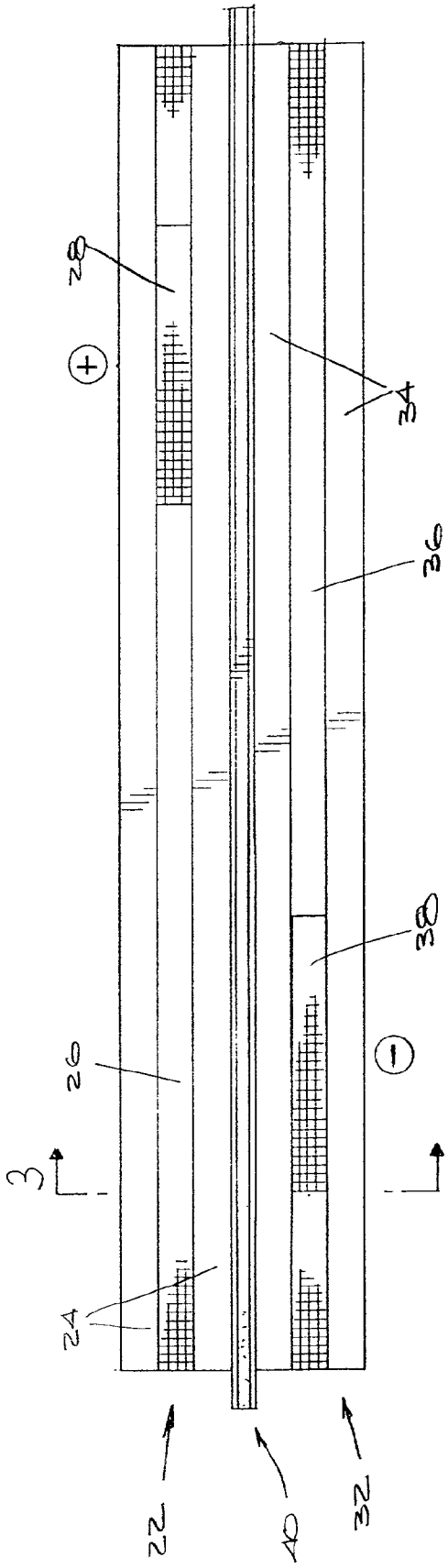


FIG. 2

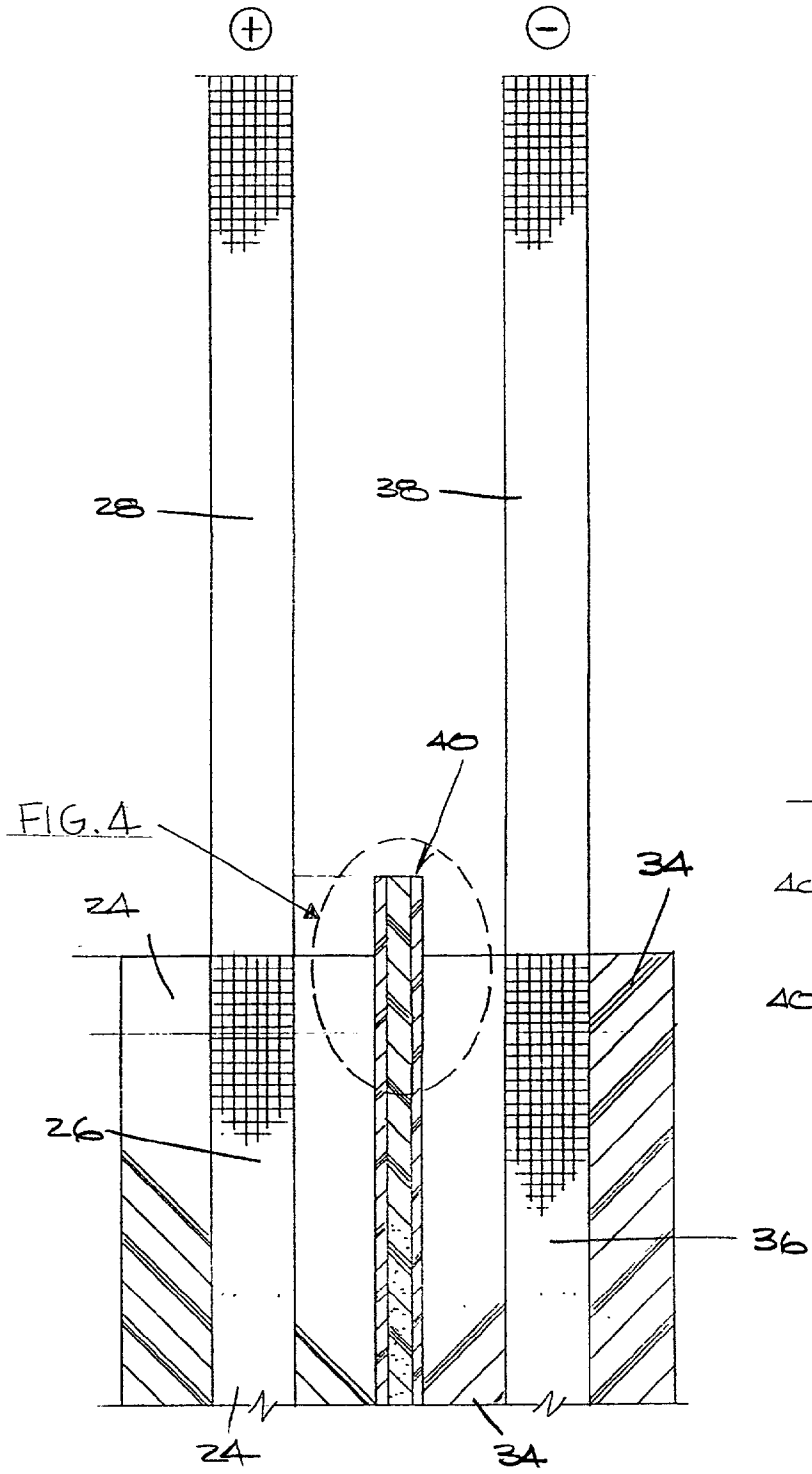


FIG. 3

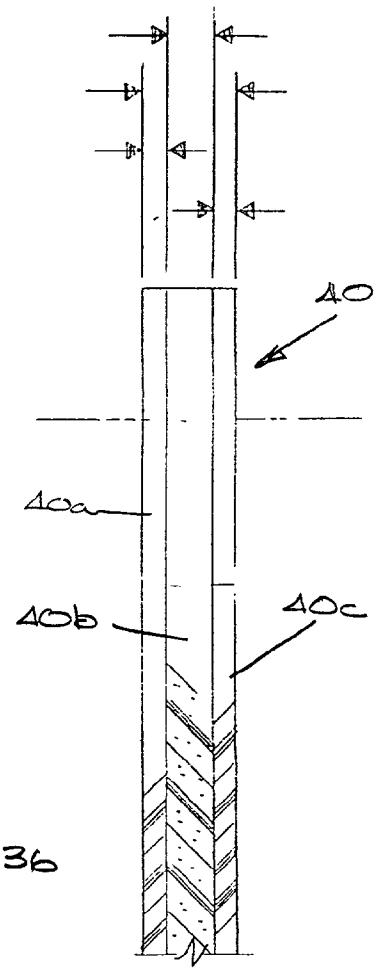
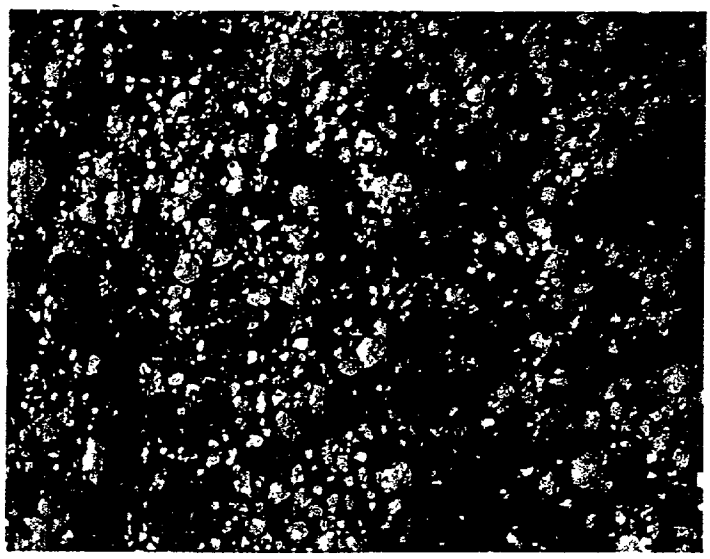
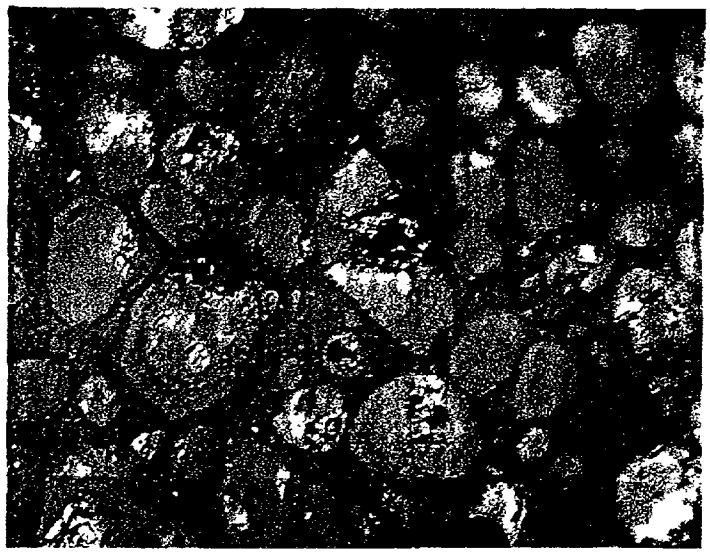


FIG. 4



X 2000

FIG. 5



X 2000

FIG. 6

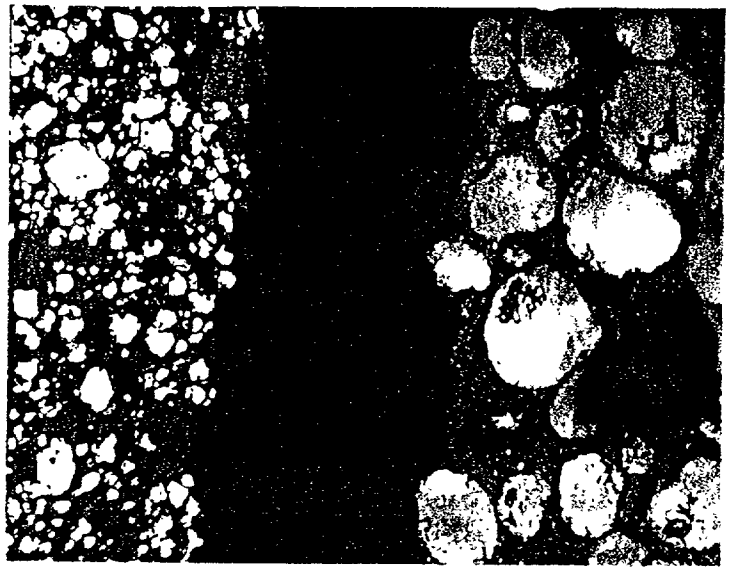


FIG. 7

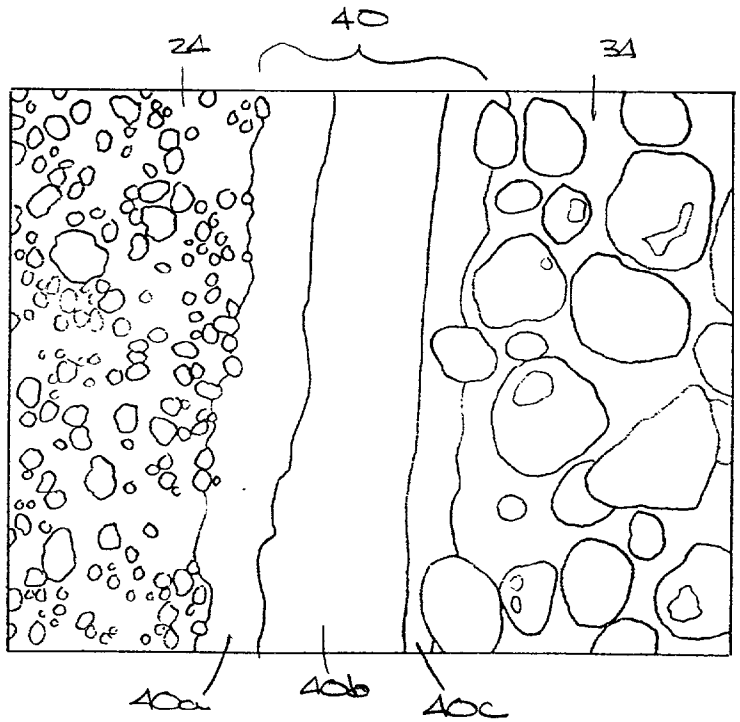


FIG. 8

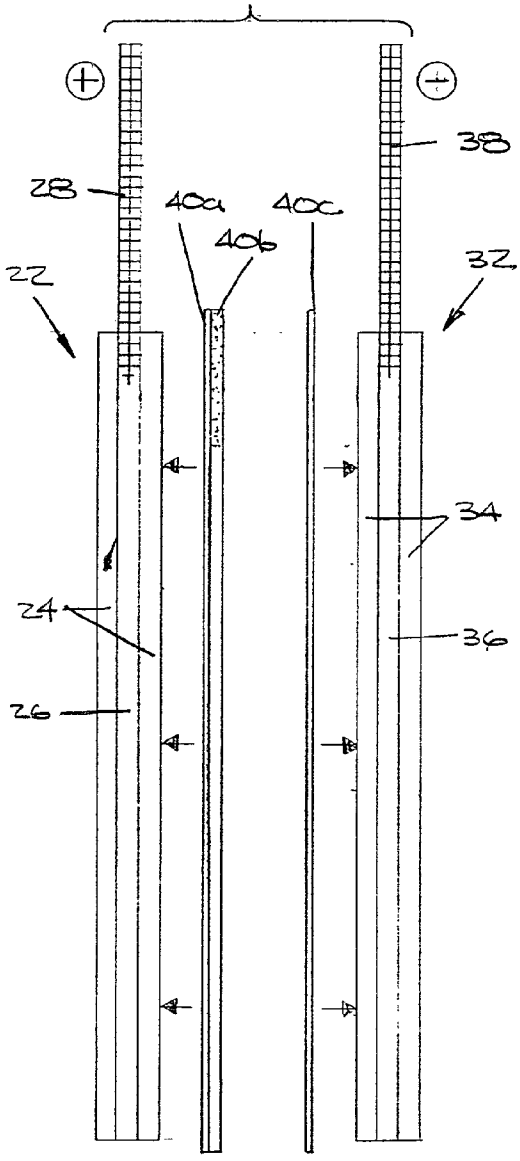

22
28
40a
40c
40b
38
32
34
24
36
26

FIG. 9A

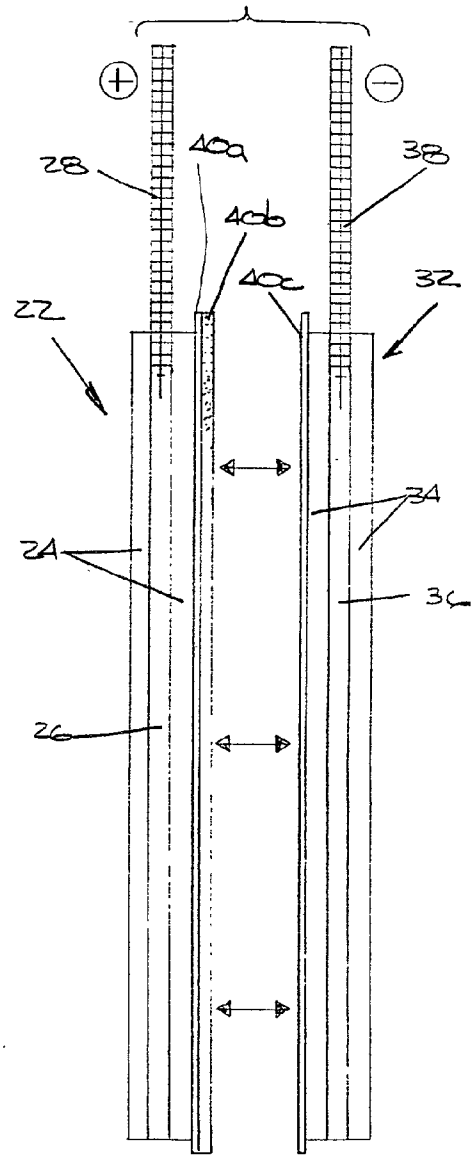

22
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40a
40b
40c
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FIG. 9B

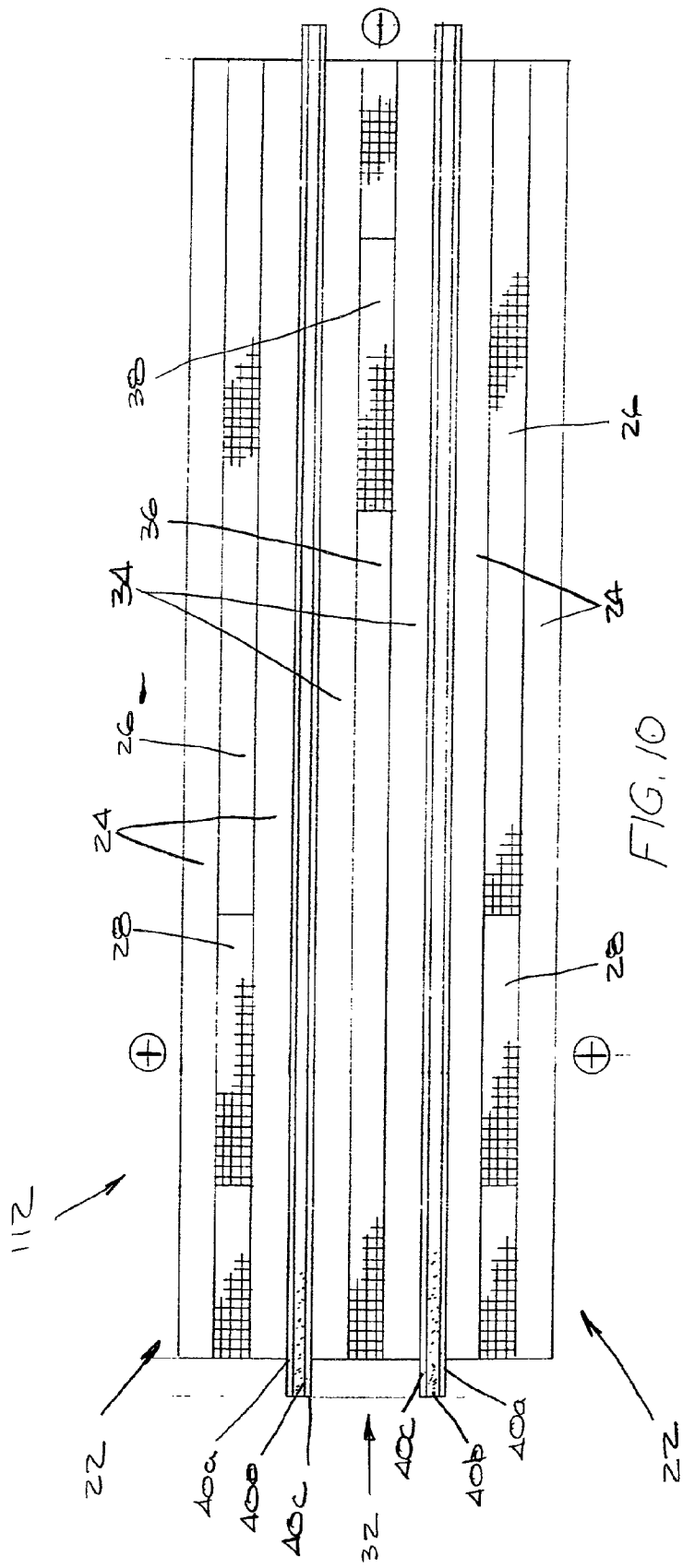


FIG. 10

BATTERY SEPARATOR FOR LI-ION AND/OR LI-ION POLYMER BATTERY

FIELD OF THE INVENTION

[0001] The present invention relates generally to batteries, fuel cells and electrochemical cells, and more particularly to separators used in such devices. The invention is particularly applicable to lithium-ion polymer batteries and will be described with respect thereto. It will be appreciated, however, that the present invention finds advantageous applications in other devices, such as those indicated above.

BACKGROUND OF THE INVENTION

[0002] Most batteries include a component referred to as a "separator" that is adapted to separate, and prevent direct contact between, an anode section and a cathode section. In addition to maintaining physical separation of the anode section and the cathode section, a separator is designed to perform several other functions such as forming an ionic pathway between the anode section and cathode section, electronic insulation, mechanical support and as a layer binding the anode section and cathode section. Each of these requirements often dictates the need for different and sometimes opposite material properties. For example, a material having high tensile strength may have poor flexing properties, or a material having a high porosity may have poor adhesive properties. Thus, the design and manufacture of a separator for use in a battery requires a balancing of a number of material properties such as mechanical strength, adhesion, porosity and processability.

[0003] The present invention provides a multi-layered separator component for use in an electromechanical device, where the multi-layered separator has at least two layers formed of different separator materials and wherein each layer has different physical properties than the other.

SUMMARY OF THE INVENTION

[0004] In accordance with a preferred embodiment of the present invention, there is provided a separator for a lithium-ion polymer battery that is comprised of a plurality of separator layers that are laminated together. The plurality of separator layers includes a first layer formed of a first separator material and a second layer formed of a second separator material, wherein the second layer is structurally and compositionally different from the first layer.

[0005] In accordance with another aspect of the present invention, there is provided a Li-ion and/or Li-ion polymer cell that is comprised of a cathode section, an anode section and a multi-layered separator disposed between the anode section and the cathode section. The multi-layered separator includes a first layer formed of a first separator material and a second layer formed of a second separator material, wherein the first layer has different physical properties than the second layer.

[0006] In accordance with another aspect of the present invention, there is provided a method of forming a Li-ion or Li-ion polymer battery having a multi-layered separator, comprising the steps of:

[0007] a) forming an anode section and a cathode section, the anode section having an anode layer adhered to an anode current collector layer and the cathode section having a cathode layer applied to a cathode current collector;

[0008] b) laminating a first separator layer formed of a first separator material to the anode layer of the anode section and laminating a second separator layer formed of a second separator material to the cathode layer of the cathode section, wherein the first separator layer is compositionally and structurally different than the second separator layer;

[0009] c) laminating the anode section to the cathode section wherein the first separator layer and the second separator layer are disposed between the anode section and the cathode section.

[0010] It is an object of the present invention to provide a separator component for an electromechanical device.

[0011] It is another object of the present invention to provide a separator component as described above, wherein the separator component is comprised of multiple layers, wherein each layer is formed of a different separator material having different physical and/or chemical properties.

[0012] Another object of the present invention is to provide a Li-ion and/or Li-ion polymer cell having a multi-layered separator as described above.

[0013] A still further object of the present invention is to provide a method of forming a multi-layered separator as part of a Li-ion and/or Li-ion polymer cell.

[0014] These and other objects will become apparent from the following description of a preferred embodiment taken together with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The invention may take physical form in certain parts and arrangement of parts, a preferred embodiment of which will be described in detail in the specification and illustrated in the accompanying drawings which form a part hereof, and wherein:

[0016] **FIG. 1** is a perspective view of a battery cell having a multi-layered separator, illustrating a preferred embodiment of the present invention;

[0017] **FIG. 2** is an enlarged, top view of the battery cell shown in **FIG. 1**;

[0018] **FIG. 3** is a sectional view taken along lines 3-3 of **FIG. 2** showing the upper end of the battery cell shown in **FIG. 1**;

[0019] **FIG. 4** is an enlarged view of area 4-4 in **FIG. 3**;

[0020] **FIG. 5** is a micrograph at 2,000×magnification showing the structure of a cathode;

[0021] **FIG. 6** is a micrograph at 2,000×magnification showing the structure of an anode;

[0022] **FIG. 7** is a micrograph at 2,000×magnification showing a three-layered separator between a cathode and an anode, according to the present invention;

[0023] **FIG. 8** is a mechanical representation of the three-layered separator shown in **FIG. 7**;

[0024] **FIGS. 9A and 9B** are schematic views showing a preferred method of forming a battery cell having a multi-layered separator, in accordance with one aspect of the present invention; and

[0025] **FIG. 10** is a bi-cell having two multi-layered separators illustrating another embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

[0026] Referring now to the drawings wherein the showings are for the purpose of illustrating a preferred embodiment of the invention only, and not for the purpose of limiting same, **FIG. 1** shows a battery **10** illustrating a preferred embodiment of the present invention. Battery **10** is formed of a polymer, electrolytic cell **12** contained within a package **14** (shown in phantom in the drawings) that is formed of a flexible laminate material. Cell **12** is preferably a lithium-based electrochemical cell. Cell **12** may be a primary (non-rechargeable) cell or a secondary (rechargeable) cell. Cell **12** is comprised of a cathode section **22** and an anode section **32**.

[0027] Cathode section **22** is comprised of two layers **24** of a cathode film. The film-forming cathode layer **24** is preferably comprised of a lithiated metal oxide active material, a conductive material and a binder material. A current collector **26** formed of a metal screen, metal mesh or a sheet of perforated metal is provided between cathode layers **24**. Current collector **26** preferably has a thickness of about 25 μm to about 50 μm. Current collector **26** includes an outward extending tab or strip **28**. Each cathode layer **24** preferably has a thickness of about 50 μm to about 200 μm, and more preferably about 80 μm to about 150 μm. **FIG. 5** is a sectional view of cathode film layer **24** at 2,000×magnification showing lithiated metal oxide particles and conductive particles (the lighter areas) in a binder matrix (the darker background).

[0028] Anode section **32** is comprised of two layers **34** of an anode film having a current collector **36** disposed therebetween. Current collector **36** is preferably formed of a metal mesh, metal screen or a sheet of perforated metal having a thickness of about 25 μm to about 50 μm. The film-forming anode layers **34** are preferably comprised of a carbon active material, a conductive material and a binder material. Current collector **36** includes an outward extending tab or strip **38** that defines the negative lead of battery **10**. Each anode layer **34** preferably has a thickness of about 50 μm to about 200 μm, and more preferably about 80 μm to about 150 μm. **FIG. 6** is a sectional view of anode film layer **34** at 2,000×magnification showing the carbon active material (the larger, light areas) in a binder matrix (the darker background area).

[0029] Between anode section **32** and cathode section **22**, a separator **40** is disposed. In accordance with the present invention, separator **40** is a multi-layered, laminated structure having two or more separator layers, designated **40***a*, **40***b* and **40***c* in the drawings. In the embodiment shown, separator **40** is comprised of a first separator layer **40***a*, a second separator layer **40***b* and a third separator layer **40***c*. The overall thickness of cell **12** is about 800 μm or less, and preferably about 500 μm or less.

[0030] Each layer **40***a*, **40***b* and **40***c* is formed of a separator material. Such material may include, by way of example and not limitation, the following: a polymer, a plasticizer and a filler. The separator material forming a particular separator layer **40***a*, **40***b* and **40***c* is selected to provide such layer with specific physical and structural properties.

[0031] Each separator layer is preferably formed of a specific, and different, separator material. As used herein, the term "separator material" refers to materials suitable for forming a separator layer. In the context of the present invention, two separator layers are considered to be formed from "different separator materials" if they are formed of different constituent components, or if they are formed from the same constituent components, but such components are present in different weight percentages or ratios. If separator **40** is comprised of three or more separator layers **40***a*, **40***b*, **40***c* etc., two or more of the separator layers may be formed of the same separator material, however, preferably, no two adjacent separator layers are formed of the same separator material.

[0032] In the embodiment shown, separator layers **40***a*, **40***c* are essentially identical, and are formed of the same separator material. Because layers **40***a*, **40***c* abut respective cathode film **24** and anode film **34**, they are preferably formed of a separator material that has good adhesive properties. Typically, a separator layer formed of a softer and less porous material provides better adhesive properties.

[0033] Each separator layer **40***a*, **40***c* preferably has a thickness of about 10 μm to about 75 μm, and more preferably about 15 μm to about 35 μm.

[0034] Layer **40***b* is preferably formed to be relatively harder and more rigid than separator layers **40***a*, **40***c*, so as to provide structural support. As shown in the drawings, separator layer **40***b* is relatively thicker than separator layers **40***a*, **40***c*. Layer **40***b* preferably has a thickness of about 20 μm to about 100 μm, and more preferably about 30 μm to about 60 μm.

[0035] Layers **40***a*, **40***b* and **40***c* are preferably formed of conventional separator material. A typical formulation for a separator includes a polymer, a filler and a plasticizer pore former (which is later removed). Separator layer **40***b* may have a filler composition ranging from 22% by weight (the amount typically used in forming conventional separators), to a maximum percentage that can be processed into a polymer. The preferred filler range is about 25% to about 50%, and most preferably 30% to 40% of the separator material. This filler composition may be combined with any combination of polymer, plasticizer, etc. required to give an appropriate combination of properties for a given application (i.e., battery performance). In general, preferred ranges of polymer are from 20% to 40%, and plasticizer from 30% to 50%.

[0036] The filler itself may compromise more than one type of material, and again a combination may be chosen to maximize a desired property, e.g., density. In general, the largest average particle diameter chosen will not exceed half the thickness of the layer in which it is incorporated. The preferred filler is silica.

[0037] The overall thickness of the separator may vary from less than 10 μm to 1 mm depending on the device and

the application. For the specific example given, a preferred range is 30 to 80 μm for final thickness of the graded separator.

[0038] A separator layer is formed by dissolving a polymer in an appropriate organic solvent (e.g., acetone) followed by thoroughly dispersing the filler in the polymer solution. Once the filler is dispersed in the polymer solution, a plasticizer is added. The slurry so formed is then cast into a thin film on a carrier by conventional film casting techniques. The cast, but slightly poor separator film, may be subsequently densified and laminated using a hot roll laminator to produce a strong, flexible film.

[0039] As will be appreciated by those skilled in the art, various materials, compositions and formulas may be used to produce a separator film. By varying the materials and composition, separator films having a wide difference in properties may be formed.

[0040] The present invention provides a multi-layered separator **40** that has at least two separator layers **40***a*, **40***c*, each having physical properties and a structural make-up that is different from the other separator layer **40***b*. **FIG. 7** is a sectional view at 2,000×magnification, showing a three-layered separator **40** disposed between a cathode film **24** and an anode film **34**. **FIG. 8** is a mechanical representation of the image shown in **FIG. 7** that is provided for easier identification of the respective layers and components. As shown in **FIGS. 7 and 8**, separator layers **40***a*, **40***c* are relatively thinner as contrasted with intermediate separator layer **40***b*. As indicated above, layers **40***a*, **40***c* are preferably formed of a composition that has low porosity and improved adhesive properties. The improved adhesive properties are particularly important in providing adhesion to anode film layer **34**. As seen in **FIGS. 7 and 8**, the relatively large carbon particles of film **34** produce a rougher exterior surface. A porous, relatively hard separator film would not have good adhesion to such a surface. By providing separator layer **40***c* that is a relatively soft and less porous material, separator layer **40***c* can more easily mate with the surface of anode film **34** and provide better adhesion thereto. Similarly, separator layer **40***a* provides better adhesion to cathode film **24**.

[0041] The present invention shall now be described by way of the following Example.

EXAMPLE

[0042] The following Example relates to the construction of a Li-ion polymer battery using inert matrix technology. In this type of construction, a common polymeric binder, typically a polyvinylidene fluoride (PVDF) co-polymer, is used to form the three basic component layers of the battery, i.e., cathode films **24**, anode films **34** and separator **40**, i.e., separator layers **40***a*, **40***b* and **40***c*. Separator layers **40***a*, **40***b* and **40***c* are ultimately combined, typically by heat and pressure, into a monolithic porous laminate, into which electrolyte solution can be infused to create a battery. As indicated above, a separator used in such a construction will comprise a polymer (which acts as the binder or matrix), a plasticizer pore former (which is removed later), and a filler. The composition used is chosen to provide a balance of mechanical strength, adhesion, final porosity and processability.

[0043] Separator **40** comprises three layers **40***a*, **40***b* and **40***c*. The two outer layers **40***a*, **40***c* have the following composition:

Component	Material	Weight Percentage
Polymer	Polyvinylidene fluoride	33.4
Plasticizer	dibutyl phthalate	44.4
Filler	silica powder (Type I)	22.2

[0044] Inner layer **40***b* differs in that it has a higher proportion of filler. Inner separator layer **40***b* is prepared with the following composition:

Component	Material	Weight Percentage
Polymer	polyvinylidene fluoride	30
Plasticizer	dibutyl phthalate	35
Filler	silica powder (Type I)	22
Filler	silica powder (Type II)	13

[0045] The respective separator layers **40***a*, **40***b* and **40***c* are formed from a slurry, as previously described. In this respect, a slurry is prepared by first dissolving the polyvinylidene fluoride polymer in an appropriate organic solvent (e.g., acetone), followed by thoroughly dispersing the silica powder in the polymer solution. Once the silica is dispersed in the polymer solution the plasticizer is added. Then the slurry is cast into a thin film by conventional film casting techniques on a carrier. The cast, but slightly porous separator film, can be subsequently densified and laminated using a hot roll laminator to produce a strong, flexible film.

[0046] Separator layers **40***a*, **40***b* and **40***c* are combined, together with cathode and anode sections **22**, **32**, by a sequence of roll laminations at elevated temperatures and pressures, into a monolithic battery structure.

[0047] Separators made from multiple, laminated separator layers **40***a*, **40***b* and **40***c* may vary from 15 μm and up in thickness and be combined into a final separator **40** with thicknesses ranging from 30 μm to 100 μm.

[0048] A multi-layer separator **40**, as described above, has greater hardness and greater compressive strength than typical, conventional single layer separators. This increases its resistance to penetration and shorting.

[0049] In accordance with the present invention, methods and techniques for forming separator **40** are very broad. The different separator layers **40***a*, **40***b* and **40***c* combined into separator **40** may comprise different materials—any effective polymer (including copolymers and blends), filler, plasticizer or pore former including various compositions thereof may be used. Separator layers **40***a*, **40***b* and **40***c* may be combined in a number of ways known to those skilled in the art. These include separate construction of films and their subsequent combination by roll or press lamination, or by adhesive bonding. Alternatively, separator layers **40***a*, **40***b* and **40***c* may be built up by sequential film-forming processes, e.g., by sequential casting, printing, coating or spraying processes, or by some combination of these. Combina-

tions of any and all of these processes may be possible, depending on the number and nature of the layers desired in final separator **40**.

Method of Forming a Battery

[**0050**] A multi-layered separator **40**, made in accordance with the present invention, may be formed as a separate component, wherein separator layers **40***a*, **40***b* and **40***c* are individually formed and then joined into multi-layered separator **40** that is then laminated to cathode section **22** and anode section **32**. In accordance with a preferred method of forming a battery **10**, separator layers **40***a* and **40***c* are preferably first adhered, separately, to cathode film layer **24** and anode film layer **34**, respectively, prior to their joining with intermediate separator layer **40***b*. In this respect, **FIG. 9** schematically illustrates a preferred method of forming separator **40**.

[**0051**] In accordance with a preferred method of forming a battery, it is preferable that the softer, less porous outer separator layers **40***a*, **40***c* are first laminated to their respective cathode film layer **24** and anode film layer **34** and then cathode section **22** with separator layer **40***a* thereon and anode section **32** with separator layer **40***c* thereon are then laminated together with intermediate separator layer **40***b* disposed between separator layers **40***a* and **40***c*. It will, of course, be appreciated by those skilled in the art that since cathode film layer **24** is relatively smooth, separator layer **40***a* and separator layer **40***b* may be applied simultaneously to cathode film **24** of cathode section **22** with satisfactory results.

[**0052**] More specifically, separator layer **40***a* is first joined to cathode section **22** by adhering separator layer **40***a* to cathode film **24**. Similarly, separator layer **40***c* is preferably first adhered to anode film **34** of anode section **32**. By laminating separator layer **40***c* directly to anode film layer **34** of anode section **32**, more uniform laminating pressure may be exerted upon separator layer **40***c* since the surface applying the mechanical pressing (not shown) engages separator layer **40***c* directly and thus the laminating pressure is not dissipated by any intermediate layer that may be disposed between the mechanical surface and separator layer **40***c*. This facilitates better adhesion of separator layer **40***c* to the relatively rougher surface of anode film layer **34**. Since separator layer **40***c* has a similar composition to separator layer **40***b*, laminating these layers to each other in a subsequent step still provides good adhesion to form the multi-layered separator layer **40**.

[**0053**] The foregoing description of a preferred method of forming multi-layered separator **40** shows how the respective layers may be applied to cathode section **22** and anode section **32** based upon the surface conditions and adhesion characteristics of the respective film layers and separator layers. In accordance with the preferred method of the present invention, multi-layered separator **40** is preferably formed by first adhering one or more of the separator layers to a cathode film layer **24** of cathode section **22** and to an anode film layer **34** of anode section **32** and then subsequently joining the respective separator layers together to form the multi-layered separator **40**.

[**0054**] Referring now to **FIG. 10, a** bi-cell **112** illustrating an alternate embodiment of the present invention is shown. As contrasted with cell **112** illustrated in FIGS. **1-9**, bi-cell **112** has two cathode sections **22** and an anode section **32** disposed therebetween. Cathode sections **22** and anode sections **32** are similar to those heretofore described, and like reference numbers are used to identify like components. Disposed between each cathode section **22** and anode section **32** is a multi-layered separator **40** similar to that heretofore described. **FIG. 10** thus illustrates how a multi-layered separator **40** in accordance with the present invention finds advantageous application in multi-cell configurations.

[**0055**] The present invention thus provides a multi-layered battery separator for a lithium-ion and/or lithium-ion polymer battery, fuel cells and electrochemical cells. The present invention provides a means of increasing the range of requirements that can be met by combining separator layers that have different structural and physical properties. In a final battery construction, separator layer **40** will be a monolithic laminate (itself incorporated into the monolithic structure of the whole battery) comprising separator layers **40***a*, **40***b*, **40***c* etc. of different composition and hence properties. By associating different properties with different separator layers **40***a*, **40***b*, **40***c* in separator **40**, the present invention advantageously separates the requirements and reduces the compromises that are usually necessary if a single separator composition is used. Because different materials and compositions can all be incorporated into a final, single separator construction, the range of properties that can be provided is increased. It is believed that separator **40**, in accordance with the present invention, can translate into less expensive material or manufacturing costs, easier manufacturing processes, higher yields, better battery performance, greater abuse resistance of the battery and an enhanced range of operating conditions.

[**0056**] The foregoing description is a specific embodiment of the present invention. It should be appreciated that this embodiment is described for purposes of illustration only, and that numerous alterations and modifications may be practiced by those skilled in the art without departing from the spirit and scope of the invention. It is intended that all such modifications and alterations be included insofar as they come within the scope of the invention as claimed or the equivalents thereof.

Having described the invention, the following is claimed:

1. A multi-layered separator for a Li-ion polymer battery, comprised of:

a plurality of separator layers that are laminated together, said plurality of separator layers including a first layer formed of a first separator material and a second layer formed of a second separator material, wherein the second layer is compositionally and structurally different from the first layer.

2. A multi-layered separator as defined in claim 1, wherein said first layer is adjacent to said second layer.

3. A multi-layered separator as defined in claim 2, wherein said separator includes two separator layers.

4. A multi-layered separator as defined in claim 1, wherein said separator includes a third layer formed of said first separator material, and said layers are arranged wherein said second layer is disposed between said first and third layer.

5. A multi-layered separator as defined in claim 4, wherein said first layer and said third layer have better adhesive properties than said second layer.

6. A multi-layered separator as defined in claim 1, wherein each of said separator layers is formed from a composition including a polymer, a filler and a plasticizer.

7. A multi-layered separator as defined in claim 6, wherein said polymer is polyvinylidene fluoride, said filler is silica and said plasticizer is dibutyl phthalate.

8. A multi-layered separator as defined in claim 6, wherein said first layer and said second layer are formed from the following:

about 33.4% by weight polyvinylidene fluoride,

about 44.4% by weight dibutyl phthalate and

about 22.2% by weight silica powder, and said second layer is comprised of the following:

about 30% by weight polyvinylidene fluoride,

about 35% by weight dibutyl phthalate and

about 35% by weight silica powder.

9. A Li-ion and/or Li-ion polymer cell, comprised of:

a cathode section;

an anode section; and

a multi-layered separator disposed between said anode section and said cathode section, said separator including a first layer formed of a first separator material and a second layer formed of a second separator material, wherein said first layer has different physical properties than said second layer.

10. A Li-ion and/or Li-ion polymer cell as defined in claim 9, wherein said separator includes a third layer formed of said first separator material, and said layers are arranged wherein said second layer is disposed between said first and third layer.

11. A Li-ion and/or Li-ion polymer cell as defined in claim 9, wherein said first layer is adjacent to said second layer.

12. A Li-ion and/or Li-ion polymer cell as defined in claim 9, wherein said separator includes two separator layers.

13. A Li-ion and/or Li-ion polymer cell as defined in claim 9, wherein each of said separator layers is formed from a composition including a polymer, a filler and a plasticizer.

14. A Li-ion and/or Li-ion polymer cell as defined in claim 13, wherein said polymer is polyvinylidene fluoride, said filler is silica and said plasticizer is dibutyl phthalate.

15. A method of forming a Li-ion or Li-ion polymer battery having a multi-layered separator, comprising the steps of:

a) forming an anode section and a cathode section, said anode section having an anode layer adhered to an anode current collector layer and said cathode section having a cathode layer applied to a cathode current collector;

b) laminating a first separator layer formed of a first separator material to said anode layer of said anode section and laminating a second separator layer formed of a second separator material to said cathode layer of said cathode section, said second separator material being different from said first separator material;

c) laminating said anode section to said cathode section wherein said first separator layer and said second separator layer are disposed between said anode section and said cathode section.

16. A method as defined in claim 15, wherein said multi-layered separator has two separator layers and said first separator layer is joined to said second separator layer.

17. A method as defined in claim 15, wherein said multi-layered separator has three layers and a third separator layer is disposed between said first separator layer and said second separator layer and is joined thereto.

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